

A Moment Classification of Exact Dyadic Causal Overhead

Bounded, logarithmic, and square-root regimes at equal entropy

A constructive note for the history-indexed C3-C model

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Abstract

We classify the order of the exact causal seed overhead for all pairs of finite pure dyadic laws with equal entropy. Unequal information variances give order $\sqrt{n}$; equal variances but unequal third centered information moments give order $\log n$; matching both moments gives bounded overhead, with a positive limit unless the laws are permutations of one another. An explicit pair on 79 and 80 symbols realizes the logarithmic regime. The proof combines an exact optimal CDF recursion, a Fourier lower bound on every entropy increment, and a smoothed envelope contraction. The lower bounds also apply to arbitrary finite action laws with equal entropy.

1 The classification

All information quantities use base-two logarithms. For a finite law A , write $\imath_A(x) = -\log_2 A(x)$ on its positive support. Define

$$h_A = \mathbb{E}\imath_A(X), \quad v_A = \mathbb{E}(\imath_A(X) - h_A)^2, \quad \kappa_A = \mathbb{E}(\imath_A(X) - h_A)^3, \quad X \sim A.$$

A *pure dyadic law* has each positive atom equal to 2^{-k} for an integer $k \geq 0$. The restriction is to individual powers of two, not to arbitrary rational numbers with a power-of-two denominator.

In the history-indexed model C3-C, one initial discrete seed determines all responses. At every positive-probability history, and under every adaptive action policy, the next output must have exactly the selected law P or Q . The policy observes past actions and outputs, not the seed. Let $C_n^{\text{causal}}(P, Q)$ be the infimum seed entropy at horizon n . The seed is allowed to depend on n .

Theorem 1 (Complete equal-entropy classification for pure dyadic laws). *Let P, Q be finite pure dyadic laws with $h_P = h_Q = h$, and set $\Delta_n = C_n^{\text{causal}}(P, Q) - nh$. The optimum is attained by an explicit finite seed. The following cases are exhaustive:*

$v_P \neq v_Q$	$: \Delta_n = \Theta(\sqrt{n}),$
$v_P = v_Q, \kappa_P \neq \kappa_Q$	$: \Delta_n = \Theta(\log(n+1)),$
$v_P = v_Q, \kappa_P = \kappa_Q$	$: \Delta_n = O(1).$

In the bounded case, Δ_n is nondecreasing and converges to a finite limit. This limit is positive unless P, Q are permutations of one another; in that exceptional case $\Delta_n = 0$ for all n . All asymptotic constants may depend on the fixed pair.

Thus, within this class, the second and third centered information moments completely determine the growth order. Higher moments may affect the bounded limit but do not create further unbounded orders.

Corollary 2 (Necessary conditions without dyadic assumptions). *For arbitrary finite action laws of equal entropy, unequal information variances imply $\Delta_n = \Omega(\sqrt{n})$. Equal variances and unequal third centered information moments imply $\Delta_n = \Omega(\log(n+1))$. In particular, bounded overhead requires matching both moments.*

2 Exact profile recursion and dyadic realization

For any discrete law A , let $F_A(t) = \mathbb{P}\{\iota_A(X) \leq t\}$, and let $\mu_A = \text{Law}(\iota_A(X))$. Define CDFs recursively by

$$F_0(t) = \mathbf{1}_{\{t \geq 0\}}, \quad F_n(t) = \min\{(\mu_P * F_{n-1})(t), (\mu_Q * F_{n-1})(t)\}. \quad (1)$$

Convolution means $(\mu * F)(t) = \int F(t - x) \mu(dx)$. The minimum of two finite-support CDFs is again a finite-support CDF.

Lemma 3 (Profile lower bound and exact dyadic attainment). *For any finite P, Q , every admissible discrete horizon- n seed S satisfies $F_S(t) \leq F_n(t)$ for all real t . For pure dyadic P, Q , F_n is the information CDF of a constructible finite pure dyadic seed R_n and $C_n^{\text{causal}} = H(R_n)$.*

Proof. At horizon zero every seed has nonnegative information. Suppose the profile bound holds for $n - 1$ and fix the first action a with output Y . For every y of positive probability, $S \mid Y = y$ is an admissible seed for the remaining horizon. On that fiber,

$$-\log_2 \mathbb{P}\{S = s\} = \iota_a(y) - \log_2 \mathbb{P}\{S = s \mid Y = y\}.$$

Induction gives $F_S(t) \leq (\mu_a * F_{n-1})(t)$ for both first actions, proving the lower bound. Integrating survival functions bounds seed entropy below by the mean of F_n .

For finite pure dyadic laws R, A , a deterministic map from R to A exists if and only if $F_R(k) \leq F_A(k)$ for all integers k . Necessity follows because a source atom must fit inside its target atom. For sufficiency, assign source atoms in decreasing order of size. At mass 2^{-k} , all eligible remaining target capacities are integer multiples of 2^{-k} and total $F_A(k) - F_R(k - 1)$, enough for the current level. Thus all bins can be filled exactly.

For $F(k) = \min(F_A(k), F_B(k))$, the counts $N_k = 2^k(F(k) - F(k - 1))$ are nonnegative integers. Taking N_k atoms of mass 2^{-k} realizes F and refines both A, B . Apply this recursively to $P \otimes R_{n-1}$ and $Q \otimes R_{n-1}$, starting at a point mass R_0 . The refinement maps send the current state to an output and a new state. Conditional on that output, the new state has exactly law R_{n-1} . Induction proves correctness after every history and for every adaptive policy. The profile lower bound proves optimality, even against non-dyadic seeds. $\square$

3 The envelope and its exact entropy increments

Assume henceforth that both entropies equal h . Put $\alpha = \text{Law}(\iota_P - h)$ and $\beta = \text{Law}(\iota_Q - h)$. Both laws are centered. Let a controller choose either increment law predictably, and let S_j^π be its sum after j steps. Define

$$G_j(x) = \inf_{\pi} \mathbb{P}\{S_j^\pi \leq x\}, \quad U_j(x) = \sup_{\pi} \mathbb{P}\{S_j^\pi \leq x\}.$$

Dynamic programming, conditioning on the first increment, gives

$$G_j = \min(\alpha * G_{j-1}, \beta * G_{j-1}), \quad G_0 = \mathbf{1}_{[0, \infty)}.$$

Consequently $G_j(x) = F_j(jh + x)$. Let Z_j have CDF G_j and put $e_j = \mathbb{E}Z_j$. For all finite action laws $\Delta_j \geq e_j$; for pure dyadic laws, Lemma 3 gives equality. The two CDFs in the minimum have the same mean e_{j-1} . Therefore

$$e_{j+1} - e_j = \frac{1}{2} \|(\alpha - \beta) * G_j\|_1 \geq 0. \quad (2)$$

Indeed, use $\min(A, B) = (A + B - |A - B|)/2$ and the CDF formula for differences of means. This identity charges a nonnegative cost at every horizon.

Write $B = \max\{|x| : x \in \text{supp } \alpha \cup \text{supp } \beta\}$. If $B = 0$, both information laws coincide and the problem is trivial. Assume $B > 0$. For every policy, conditional centering and $|D_i| \leq B$ imply

$$\mathbb{P}\{S_j^\pi \geq t\}, \mathbb{P}\{S_j^\pi \leq -t\} \leq \exp\left(-\frac{t^2}{2jB^2}\right), \quad t > 0.$$

For completeness, convexity gives $\mathbb{E}[e^{sD_i} | \mathcal{F}_{i-1}] \leq \cosh(sB) \leq e^{s^2 B^2/2}$; iterate and optimize the exponential Markov bound. The estimate is uniform over policies, hence also bounds both tails of Z_j . Integrating the tails yields

$$\mathbb{E}Z_j^2 \leq 4B^2j, \quad 0 \leq e_j \leq 2B\sqrt{j}. \quad (3)$$

4 A Fourier lower bound: why a logarithm is unavoidable

Let $D = \alpha - \beta$, and use the Fourier convention $\widehat{\nu}(t) = \int e^{itx} \nu(dx)$. Suppose $d \geq 2$ is the first order at which the moments of α, β differ. Finite support and Taylor expansion imply that for some $c_0, t_0 > 0$,

$$|\widehat{D}(t)| \geq c_0|t|^d \quad (0 < |t| \leq t_0). \quad (4)$$

Choose $c = \min(t_0, 1/(2B))$ and $t_j = c/\sqrt{j+1}$. Equation (3) and $\cos u \geq 1 - u^2/2$ show

$$|\widehat{\text{Law}(Z_j)}(t_j)| \geq \Re \widehat{\text{Law}(Z_j)}(t_j) \geq 1 - \frac{1}{2}t_j^2 \mathbb{E}Z_j^2 \geq \frac{1}{2}.$$

The function $R_j = D * G_j$ is compactly supported, and its distributional derivative is $D * \text{Law}(Z_j)$. Integration by parts therefore gives

$$\|R_j\|_1 \geq |\widehat{R}_j(t_j)| = \frac{|\widehat{D}(t_j)| |\widehat{\text{Law}(Z_j)}(t_j)|}{|t_j|} \geq \frac{c_0 c^{d-1}}{2} (j+1)^{-(d-1)/2}.$$

Together with (2), this proves

$$e_n \geq \frac{c_0 c^{d-1}}{4} \sum_{j=0}^{n-1} (j+1)^{-(d-1)/2}. \quad (5)$$

For $d = 2$ this is $\Omega(\sqrt{n})$; for $d = 3$ it is $\Omega(\log(n+1))$. Since $\Delta_n \geq e_n$ without a dyadic assumption, Corollary 2 follows already. The upper estimate in (3) proves the matching square-root upper bound for pure dyadic laws.

5 An integrated envelope contraction

For the remaining upper bounds, put

$$W_j = \int_{\mathbb{R}} (U_j - G_j)(x) dx.$$

The all- β walk has mean zero and its CDF lies between the envelopes. Thus $0 \leq e_j \leq W_j$. No reflection symmetry is assumed here.

Let V be independent smoothing noise supported in $[-A, A]$, let T_r be the all- β walk, and define

$$h_r(x) = \mathbb{P}\{T_r + V \leq x\}, \quad R_r = D * h_r, \quad K_r = |R_r|, \quad a_r = \|K_r\|_1, \quad b_r = \|K'_r\|_1.$$

The noise chosen below ensures that these functions are absolutely continuous and compactly supported where needed.

Lemma 4 (Contraction). *For every $n \geq 1$,*

$$W_n \leq 2A + 2 \sum_{r=0}^{n-1} (a_r + b_r W_{n-1-r}). \quad (6)$$

If $b_ := \sum_{r \geq 0} b_r < 1/2$, then for every $N \geq 1$,*

$$\max_{0 \leq n \leq N} W_n \leq \frac{2A + 2 \sum_{r=0}^{N-1} a_r}{1 - 2b_*}. \quad (7)$$

Proof. Write G_n^V, U_n^V for the smoothed policy envelopes and W_n^V for their integrated width. Bounded support of V gives $U_n(x) \leq U_n^V(x + A)$ and $G_n(x) \geq G_n^V(x - A)$. Every CDF F satisfies $\int (F(x + A) - F(x - A)) dx = 2A$ by Tonelli, so $W_n \leq W_n^V + 2A$.

For each policy, telescope $\mathbb{E}h_{n-i}(x - S_i^\pi)$ over i . Since $h_{r+1} = \beta * h_r$, conditioning on the past gives

$$\mathbb{P}\{S_n^\pi + V \leq x\} - h_n(x) = \sum_{i=1}^n \mathbb{E} \left[\mathbf{1}_{\{\text{action } i \text{ uses } \alpha\}} R_{n-i}(x - S_{i-1}^\pi) \right].$$

Its absolute value is at most $\sum_i \mathbb{E}K_{n-i}(x - S_{i-1}^\pi)$. Let B_j be the CDF of T_j . For every controlled prefix CDF F_j^π , $|F_j^\pi - B_j| \leq U_j - G_j$. Stieltjes integration by parts gives

$$\mathbb{E}K_r(x - S_j^\pi) \leq \mathbb{E}K_r(x - T_j) + \int |K'_r(x - y)|(U_j(y) - G_j(y)) dy.$$

The right side is independent of π , so taking the supremum first and then integrating in x proves

$$\int \sup_{\pi} \mathbb{E}K_r(x - S_j^\pi) dx \leq a_r + b_r W_j.$$

The smoothed envelope width is at most twice the largest absolute deviation from h_n at each threshold. Summing proves (6). Taking a maximum over $n \leq N$, with $W_0 = 0$, and moving $2b_*$ times that finite maximum to the left proves (7). $\square$

6 Derivative estimates for an arbitrary finite reference law

Assume β is nondegenerate. Choose two of its support points $x < y$, put $b = (y - x)/2 > 0$ and $\rho = 2 \min(\beta\{x\}, \beta\{y\}) > 0$. Then $\beta = \rho\eta + (1 - \rho)\zeta$, where $\eta = (\delta_x + \delta_y)/2$ and ζ is a probability law when $\rho < 1$. The case $\rho = 1$ needs no residual law. Let v_a denote the uniform density on $[-a, a]$, choose $a = Lb$ with a positive integer L , and take V to be the sum of four independent variables with density v_a . Then $A = 4a$ and, in weak-derivative notation,

$$h_r^{(q+1)} = (v'_a)^{*q} * v_a^{*(4-q)} * \beta^{*r}, \quad q = 2, 3, 4. \quad (8)$$

Here $v'_a = (\delta_{-a} - \delta_a)/(2a)$, and exponent zero means δ_0 . In particular $\|h_r^{(q+1)}\|_{\text{TV}} \leq a^{-q}$. The fourth case can be a signed measure; for densities the total variation norm agrees with the L^1 norm.

Lemma 5. *There are constants $C_q < \infty$ depending on β, q , but independent of $a = Lb$ and r , such that*

$$\|h_r^{(q+1)}\|_{\text{TV}} \leq \min\{a^{-q}, C_q(r+1)^{-q/2}\}, \quad q = 2, 3, 4. \quad (9)$$

Proof. Telescoping point masses shows $v'_a = \nu_L * v'_b$, where ν_L is uniform on $\{-a+b, -a+3b, \dots, a-b\}$. Convolution with probability measures cannot increase total variation. In the binomial mixture expansion of β^{*r} , the number K of η increments has law $\text{Bin}(r, \rho)$. For a fixed $K = k$, translation and scaling of symmetric signs give

$$\|(v'_b)^{*q} * \eta^{*k}\|_{\text{TV}} \leq b^{-q} \binom{k+q}{q}^{-1/2}.$$

To verify this inequality, let $N = k + q$ independent symmetric signs be ε_i . Up to sign and the factor b^{-q} , the signed measure is the law of their sum weighted by $\prod_{i=1}^q \varepsilon_i$. Given the sum, the conditional product equals the average of $\prod_{i \in I} \varepsilon_i$ over all q -element subsets I . These products are orthonormal, so that average has squared L^2 norm $\binom{N}{q}^{-1}$. Cauchy–Schwarz proves the asserted variation bound.

It remains to average over K . Using $\binom{k+q}{q} \geq (k+1)^q/q!$ and splitting at $K = \rho r/2$ gives

$$\mathbb{E} \binom{K+q}{q}^{-1/2} \leq \sqrt{q!} (1 + \rho r/2)^{-q/2} + e^{-\rho r/8} \leq C'_q (r+1)^{-q/2}.$$

For the exponential term, apply Markov to $e^{-(\ln 2)K}$ and use $1 - \rho/2 \leq e^{-\rho/2}$ and $\ln 2 < 3/4$. This proves the decay estimate; combine it with the direct bound a^{-q} from (8). $\square$

7 Closing the classification

For pure dyadic laws, the uncentered information laws have integer support. If their moments through order $m - 1$ coincide, their probability generating polynomials differ by a multiple of $(1 - z)^m$. Centering by the common h gives a finite signed measure Γ such that

$$D = (\delta_0 - \delta_1)^{*m} * \Gamma. \quad (10)$$

This follows because ordinary moments and falling-factorial moments span the same polynomials through that degree. The common translation by $-h$ is absorbed into Γ ; h need not be an integer.

Let $M = \|\Gamma\|_{\text{TV}}$. The m th unit finite difference is, up to translation, the m th weak derivative convolved with $\mathbf{1}_{[0,1]}^{*m}$, whose L^1 norm is one. Therefore

$$a_r \leq M \|h_r^{(m)}\|_{\text{TV}}, \quad b_r \leq M \|h_r^{(m+1)}\|_{\text{TV}}. \quad (11)$$

If $v_P = v_Q > 0$, the reference law is nondegenerate and Lemma 5 applies. For $m = 3$ or 4 , the summable sequence $MC_m(r+1)^{-m/2}$ dominates the bounds on b_r . As $a = Lb \rightarrow \infty$, their pointwise upper bound Ma^{-m} tends to zero. Dominated convergence lets us fix a finite a with $\sum b_r < 1/2$.

If variances match but third moments differ, use $m = 3$. Then $a_r \leq MC_2/(r+1)$, and (7) gives $W_n = O(\log(n+1))$. Since $\Delta_n = e_n \leq W_n$, the Fourier lower bound proves the logarithmic case.

If variances and third moments both match, use $m = 4$. Then $a_r \leq MC_3(r+1)^{-3/2}$ is summable, so (7) gives $W_n = O(1)$ and $\Delta_n = O(1)$. If the common variance is zero, both information laws are the same point mass and this conclusion holds trivially. Finally (2) proves monotonicity. When $\alpha \neq \beta$, it also gives $e_1 = \frac{1}{2} \int |F_\alpha - F_\beta| > 0$. For pure dyadic laws, equality of the information laws is equivalent to equality of all atom multiplicities, hence to being permutations of one another. In that case the recursion is the ordinary product profile and $\Delta_n = 0$. This completes Theorem 1.

8 An explicit logarithmic pair, with numerical constants

Take

$$P = (1/16, (1/32)^{\times 10}, (1/64)^{\times 12}, (1/128)^{\times 56}), \\ Q = ((1/32)^{\times 16}, (1/128)^{\times 64}).$$

Their support sizes are 79 and 80. Their centered information laws are

$$\alpha = \frac{1}{16}\delta_{-2} + \frac{5}{16}\delta_{-1} + \frac{3}{16}\delta_0 + \frac{7}{16}\delta_1, \quad \beta = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1.$$

Direct calculation gives

$$h_P = h_Q = 6, \quad v_P = v_Q = 1, \quad \kappa_P = -\frac{3}{8}, \quad \kappa_Q = 0, \quad D = \frac{1}{16}\delta_{-2} * (\delta_0 - \delta_1)^{*3}.$$

Let $H_n = \sum_{k=1}^n 1/k$ be the harmonic number.

Theorem 6 (A genuine logarithmic overhead). *For the displayed fixed pair and every $n \geq 1$,*

$$\boxed{\frac{H_n}{2048} \leq C_n^{\text{causal}}(P, Q) - 6n < \frac{128}{5} + \frac{3}{10}H_n.} \quad (12)$$

In particular, the overhead is $\Theta(\log n)$ and is unbounded. The exact one-step excess is $\Delta_1 = 1/8$.

Proof. For the lower bound, $B = 2$ in (3). Choose $t_j = 1/(4\sqrt{j+1})$, so $|\widehat{\text{Law}}(\widehat{Z_j})(t_j)| \geq 1/2$. For $0 \leq t \leq 1/4$, the inequality $2\sin(t/2) \geq t - t^3/24$ implies $|1 - e^{it}|^3 \geq t^3/2$. Thus $|\widehat{D}(t_j)| \geq t_j^3/32$, and (2) gives

$$e_{j+1} - e_j \geq \frac{1}{2} \frac{t_j^3/32}{t_j} \cdot \frac{1}{2} = \frac{1}{2048(j+1)}.$$

Summation proves the lower bound.

For the upper bound use four uniform smoothing variables on $[-2, 2]$, so $a = 2$ and $A = 8$. For the Rademacher reference, the derivative estimate has the explicit form

$$\|h_r^{(q+1)}\|_{\text{TV}} \leq \min\{2^{-q}, \binom{r+q}{q}^{-1/2}\}.$$

Equations (11) with $m = 3$ and $M = 1/16$ give

$$\sum_{r=0}^{n-1} a_r \leq \frac{\sqrt{2}}{16} H_n, \quad b_* \leq \frac{1}{16} \sum_{r \geq 0} \min\{2^{-3}, \sqrt{6}(r+1)^{-3/2}\} \leq \frac{3\sqrt[3]{6}}{32}.$$

For the last sum, bound it by $\int_0^\infty \min(2^{-3}, \sqrt{6}x^{-3/2})dx = 3\sqrt[3]{6}/2$. Therefore (7) and $e_n \leq W_n$ give

$$\Delta_n \leq \frac{16 + \frac{\sqrt{2}}{8}H_n}{1 - \frac{3\sqrt[3]{6}}{16}} < \frac{128}{5} + \frac{3}{10}H_n,$$

using $\sqrt{2} < 3/2$ and $\sqrt[3]{6} < 2$. Finally the CDF of $\alpha - \beta$ is $1/16, -2/16, 1/16$ on the unit intervals starting at $-2, -1, 0$. Equation (2) gives $\Delta_1 = (1 + 2 + 1)/32 = 1/8$. $\square$

9 Scope and reproducibility

The classification is for all finite pure dyadic pairs of equal entropy, with no symmetry assumption and no restriction on their alphabet sizes. For general finite laws the lower bounds remain valid, but this note does not supply matching upper constructions. It does not claim a full classification outside the pure dyadic class or literature priority.

The companion program `logarithmic_seed.py` constructs the exact profile of the explicit pair, verifies normalization and entropy using integer atom counts, and checks complete decoder partitions through horizon two. It also checks the moment identities, cubic factorization, and displayed harmonic bounds at finite horizons. Optional floating-point runs illustrate growth but are not part of the all-horizon proof. For example, a floating-point run gives $\Delta_{1000} \simeq 0.62701621$ and $\Delta_{10000} \simeq 0.69669408$. Slow apparent stabilization does not contradict logarithmic divergence.

Each horizon uses its own finite initial seed, with no later randomness. Expanded state spaces may be exponential in the horizon. No single projectively consistent infinite source or efficient expanded decoder is asserted. The proof is independent of numerical extrapolation.